CHAPTER

Dosing System

Antiscalant, pH correction and CIP dosing pumps

IN THIS CHAPTER

12.1 Overview and function

Dosing pumps inject antiscalant and pH-correction chemicals (NaOH, HCl, and in the pilot configuration also H_2SO_4) into the feed stream, and supply CIP (clean-in-place) chemicals for periodic cleaning. The documented pilot configuration used 5 dosing channels.

fig-dosing-skid.png

Figure 12.1. Suggested figure: photograph of the dosing skid, with each chemical tank and pump labeled (antiscalant, NaOH, HCl, CIP).

12.2 Key specifications

- 5 dosing channels (pilot configuration), each controlled via a 4–20 mA signal from ADAM I/O modules.
- Signal range: SET 1 = 4.0 mA (0 impulses/min, pump stopped), SET 2 = 20.0 mA (180 impulses/min).
- Chemicals confirmed in the pilot tag list and in Safety Data Sheets held by HydroVolta: antiscalant (AS), NaOH, HCl, NaOCl (sodium hypochlorite) and, in some configurations, H_2SO_4 as an alternative acid.
- Candidate dosing pump: **Etatron GB eOne** series — flow rate 1–30 l/h, max. pressure 20 bar, 100–250 V AC 50/60 Hz supply, max. 300 pulses/min, PVDF pump head with double ceramic ball valves and PTFE diaphragm. The **MF** (multifunction) variant adds proportional dosing, 4–20 mA control, level-switch input and failure alarms; the **PLUS** variant integrates a pH/ORP/Cl controller. TODO: confirm which variant, and which alternative (Lutz-Jesco MEMDOS Smart / EasyChlorGen also appear in HydroVolta's component library), is fitted to the standard SalinBloc™ product.

12.3 Operation

See section 5.4 in chapter 5 for the default 4–20 mA signal configuration and functional-test procedure used to verify each dosing channel. Setting a dosing pump's frequency from a 4–20 mA signal uses two settings: the **pulse amplitude** (mL per stroke, set manually at the pump) and the **pulse frequency** (set remotely as a % of the pump's maximum, via the 4–20 mA signal). Worked example from a real HydroVolta dosing calculation (antiscalant, 10 ppm target): dilution factor 1333, pulse amplitude 0.01–0.1 mL, giving a dose rate of ≈0.167 mL/min (36 mL/h).

WARNING

Never dose NaOH into feed or process water — HydroVolta's own CIP procedure documents this explicitly as a scaling risk. NaOH is only used against pure (permeate-quality) water during CIP.

A documented weekly CIP cycle (source: HydroVolta pilot CIP procedure) runs as follows:

1. **Acid step:** dose HCl to pH 2.5 in the CIP loop, circulating 30 minutes forward-flow then 30 minutes reverse-flow through the stack, using pretreated feedwater.
2. Drain the CIP loop to the concentrate tank, then flush with fresh feed water for 15–20 minutes.
3. **Hypochlorite step:** dose NaOCl to 2 ppm (effective only near pH 7–7.5 — check/correct pH before this step), circulating 30 minutes forward then 30 minutes reverse.
4. Drain 5 minutes, then flush with fresh feed water for 30 minutes.
5. A variant of this cycle that also treats the electrolyte compartment uses NaOH instead of/alongside HCl — **only ever with pure water**, per the warning above.

Typical HCl/NaOH CIP dose rates from a real pilot's dosing calculation sheet (pure, low-alkalinity water): HCl ≈5.5 mL/min at 1000 l/h CIP flow, pulse amplitude 0.1 mL, ≈31% of max. pump frequency; NaOH ≈17.4 mL/min, pulse amplitude 0.3 mL, ≈32% of max. frequency. For natural (higher-alkalinity) feed water, acid demand scales with the water's total alkalinity (TA) — a seawater-range feed (TA 25–30 °f) needs roughly 3–4× the acid dose of a low-alkalinity feed. TODO: confirm the CIP interval and dose rates that apply to the standard SalinBloc™ product; the figures above are a real, sourced pilot example, not a general specification.

12.4 Maintenance

TODO: no chemical replenishment interval, tubing/diaphragm replacement schedule, or dosing-pump calibration procedure specific to the standard product was found. The CIP cycle above ran approximately weekly in the documented pilot — use this as a starting reference for the maintenance schedule in chapter 7 until a formal interval is confirmed. Per the chemicals' Safety Data Sheets, store all dosing chemicals in their original, tightly closed containers, away from food/drink, at a recommended **15–25 °C**; keep incompatible chemicals (acids vs. NaOH vs. NaOCl) in separate secondary containment. See section 2.5 for first-aid and spill-response guidance.

12.5 Troubleshooting

Low-level alarms exist per chemical: `LL_Alarm_AS` (antiscalant), `LL_Alarm_NaOH`, `LL_Alarm_HCl`, plus per-channel `Dosing1_Alarm`...`Dosing5_Alarm`. On any dosing alarm, check chemical tank level first, then verify the 4–20 mA signal at the pump terminal using the bench-test method in chapter 13. If a CIP cycle fails to reach its target pH (2.5 for the acid step), verify the dosing pump's pulse amplitude/frequency setting against the worked example above before suspecting a pump fault.